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Abd Alkareem Alayasra et al.

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(54) **METHODS FOR AN IMPROVED RECEIVER FOR MULTI-SERVICE RADIO NETWORKS**

(71) Applicant: **Ulak Haberlesme A.S.**, Ankara (AR)

(72) Inventors: **Mus'ab Mahmoud Abd Alkareem Alayasra**, Istanbul (AR); **Ahmet Enes Duranay**, Istanbul (AR); **Huseyin Arslan**, Istanbul (AR)

(73) Assignee: **Ulak Haberlesme A.S.**, Ankara (TR)

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H04W 72/0446 (2023.01)

H04W 72/0453 (2023.01)

(52) **U.S. Cl.**

CPC **H04L 27/26025** (2021.01); **H04L 27/2607** (2013.01); **H04W 72/0446** (2013.01); **H04W 72/0453** (2013.01)

(58) **Field of Classification Search**

CPC H04W 72/0446; H04W 72/0453; H04L 27/26025; H04L 27/2607

See application file for complete search history.

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Primary Examiner — Brian D Nguyen

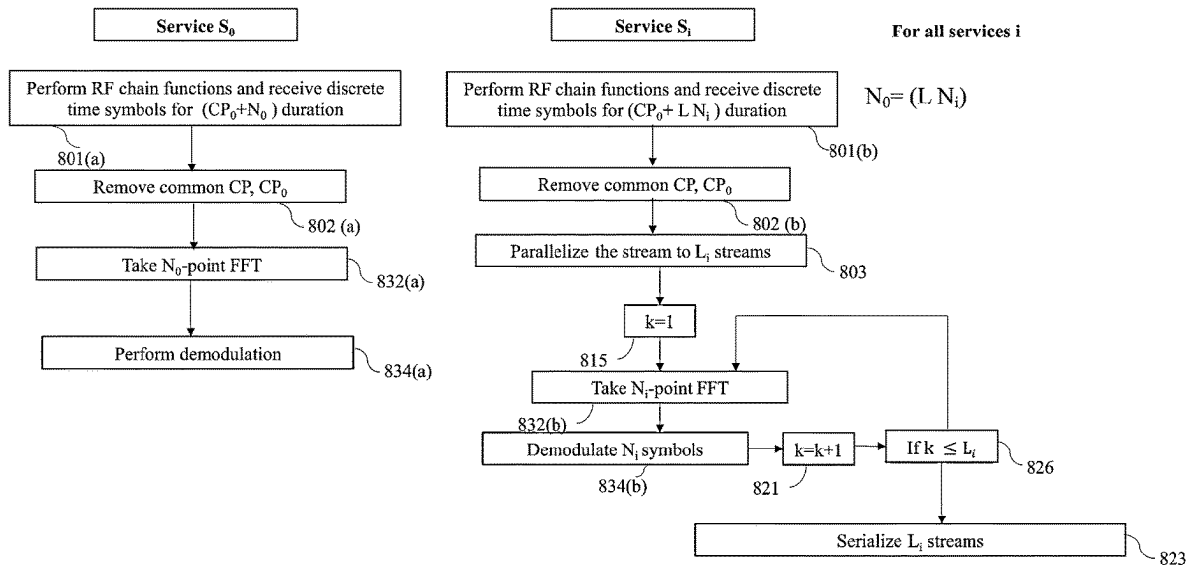
(74) *Attorney, Agent, or Firm* — Bayramoglu Law Offices LLC

(57)

ABSTRACT

A receiver of a User Equipment in a mixed numerology multi-service OFDM radio access network receives a control message for resource configuration from the base station instructing the receiver to remove a common (fixed duration) cyclic prefix (CP₀) regardless of service type when demodulating received radio signals. Upon this instruction, the receiver implements the method of removing the common cyclic prefix (CP₀) every L time domain symbols wherein $L=2^{\mu}$ for the service type that uses numerology μ . According to one embodiment, the common cyclic prefix (CP₀) duration is chosen by the base station as the cyclic prefix of the service type with longest OFDM symbol in its mix of services. The receiver eliminates inter-numerology interference (INI) by equalizing the number of subcarriers used by all service types and their CP regardless of each service type's numerology.

8 Claims, 12 Drawing Sheets



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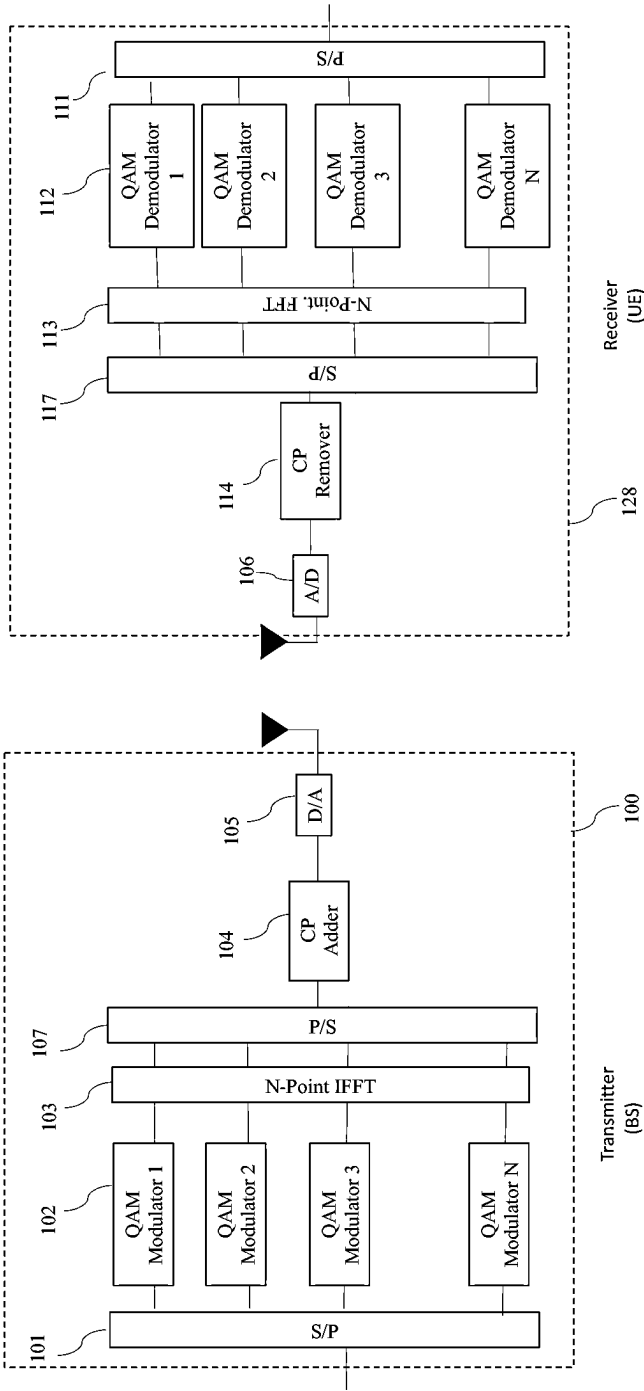


FIG. 1

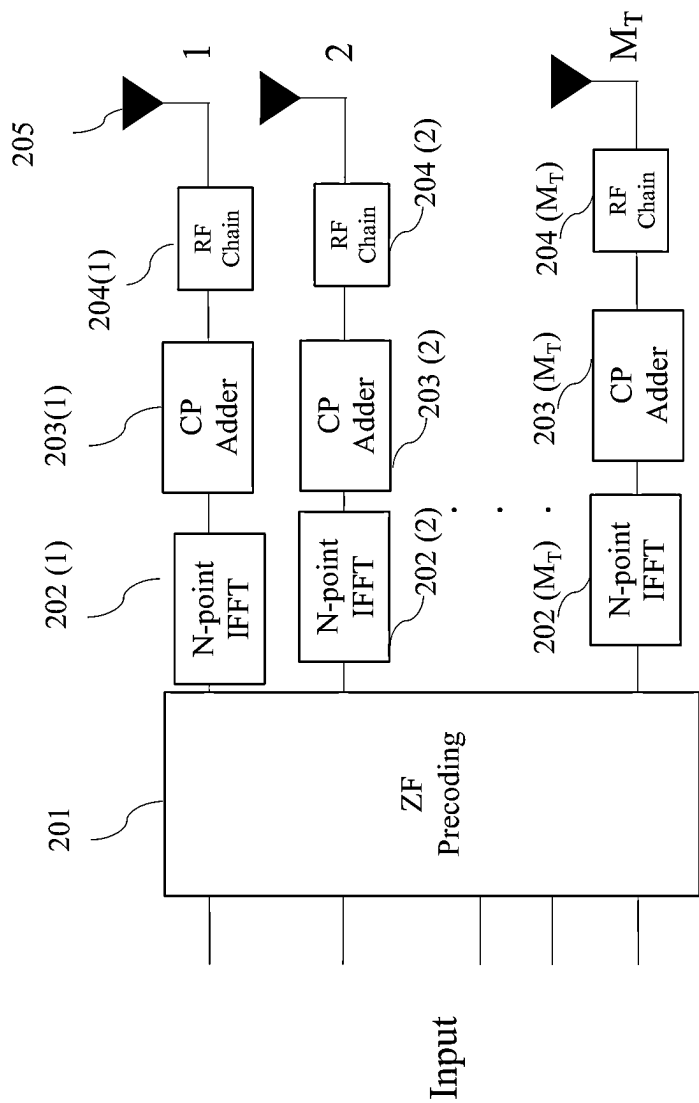


FIG. 2

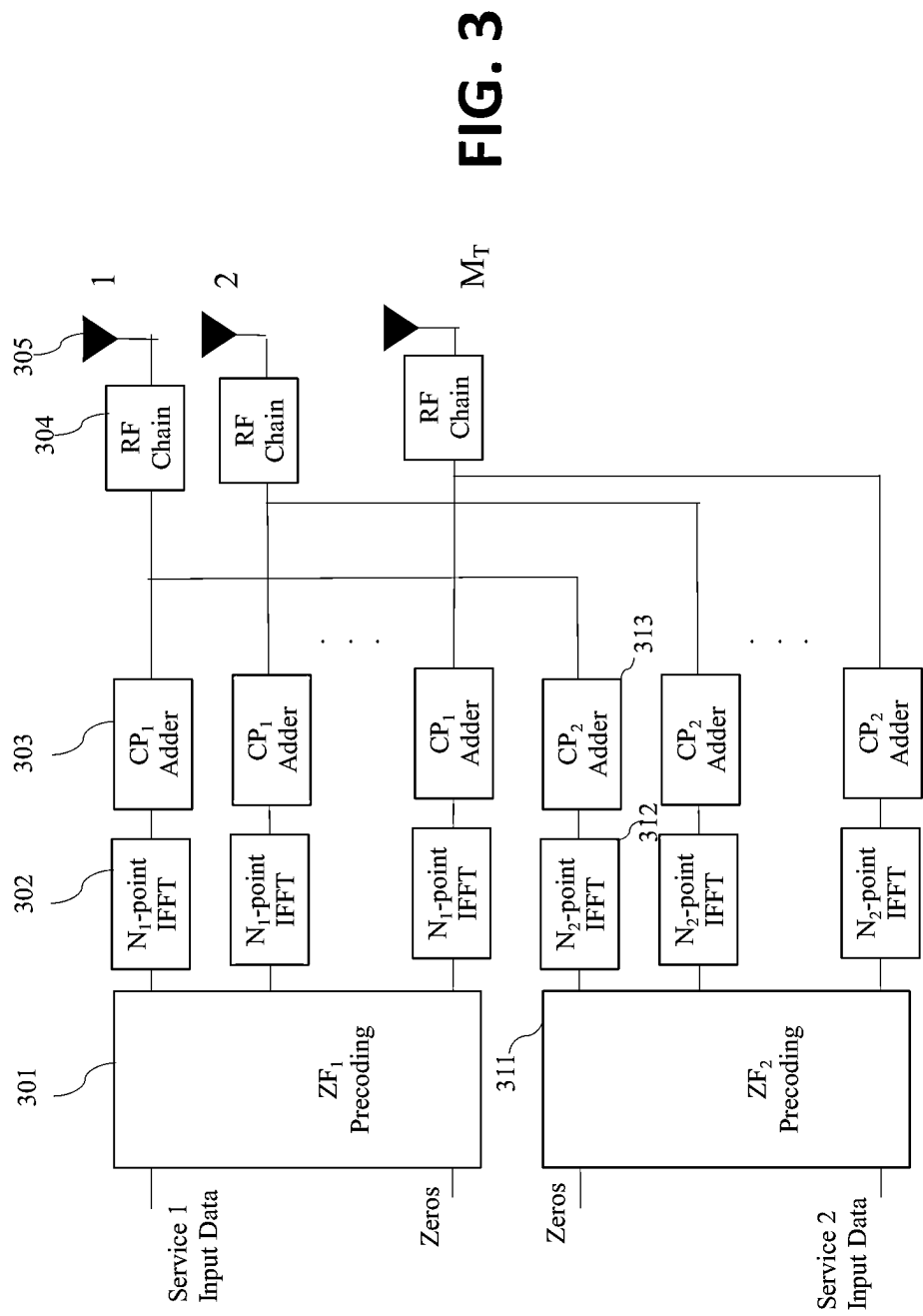


FIG. 3

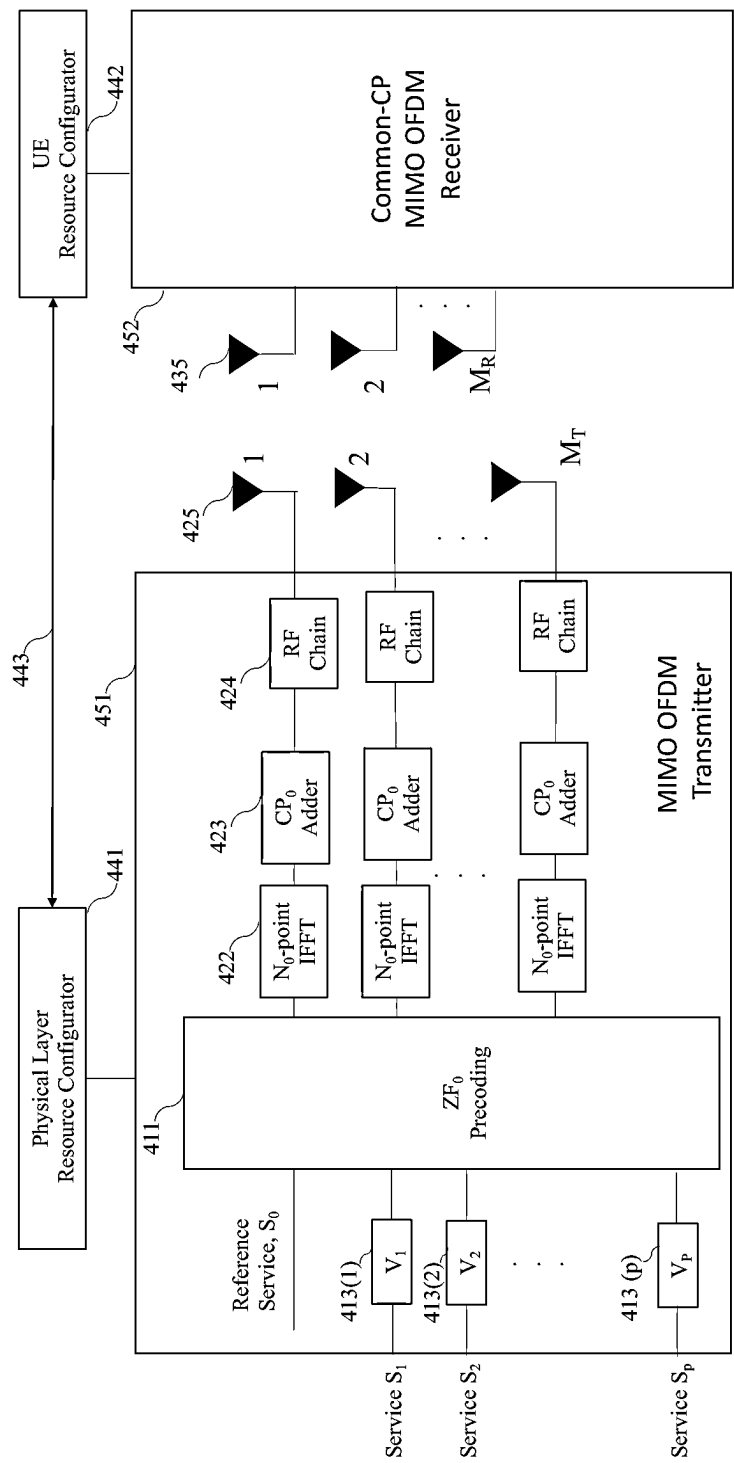


FIG. 4

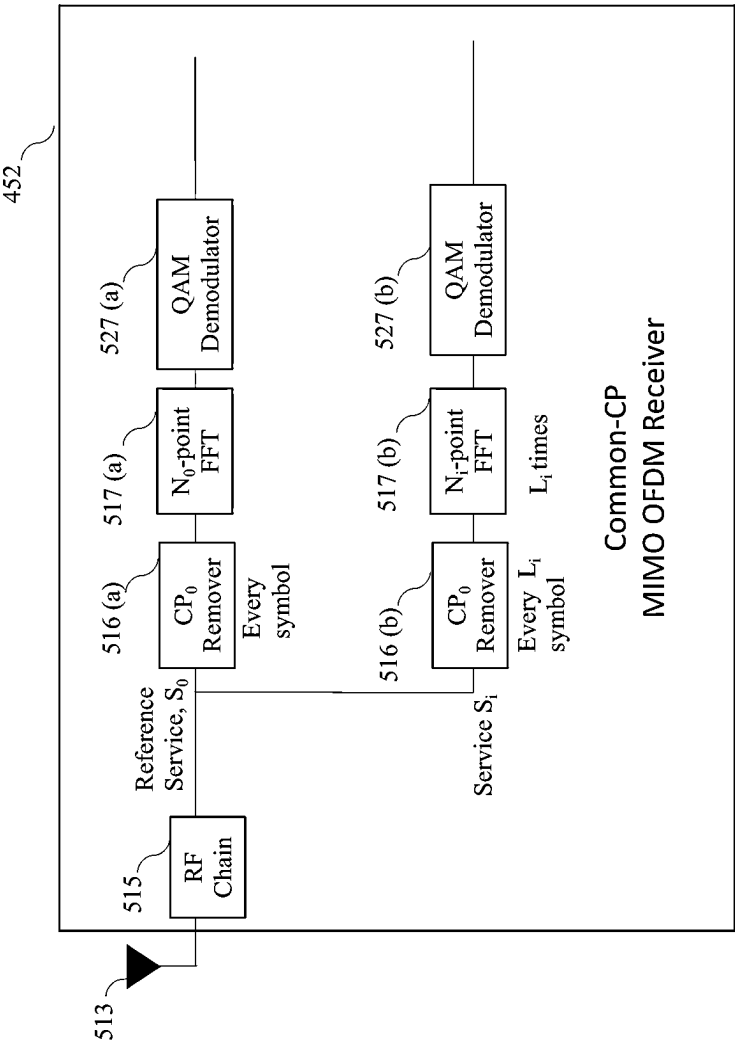


FIG. 5

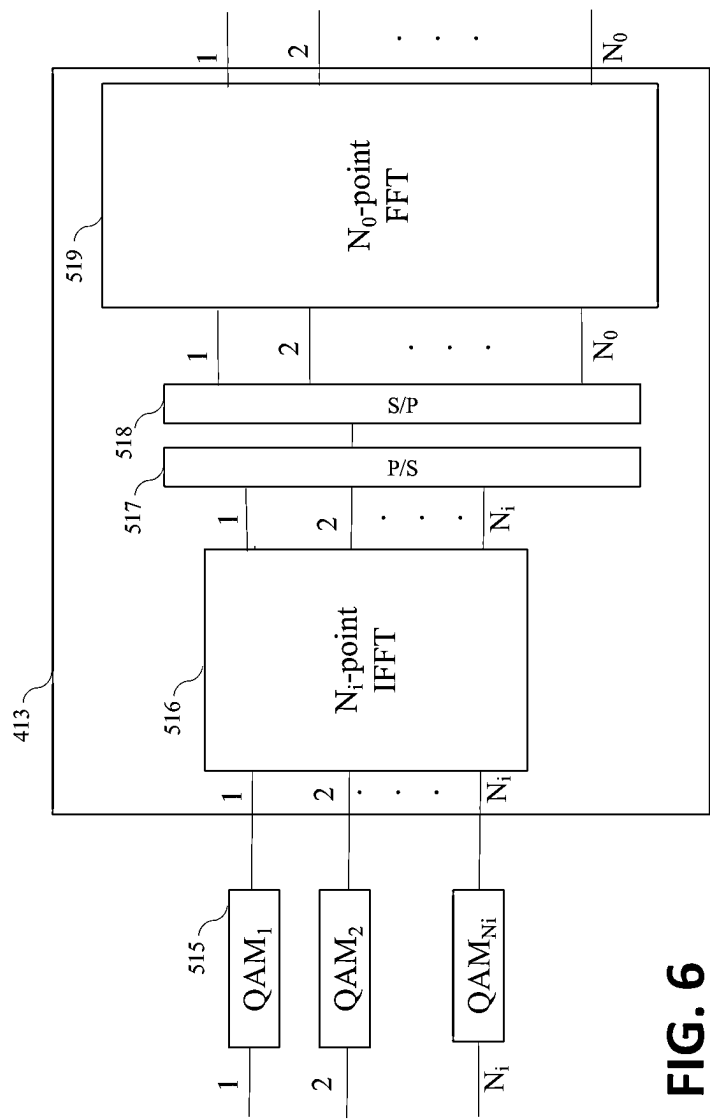
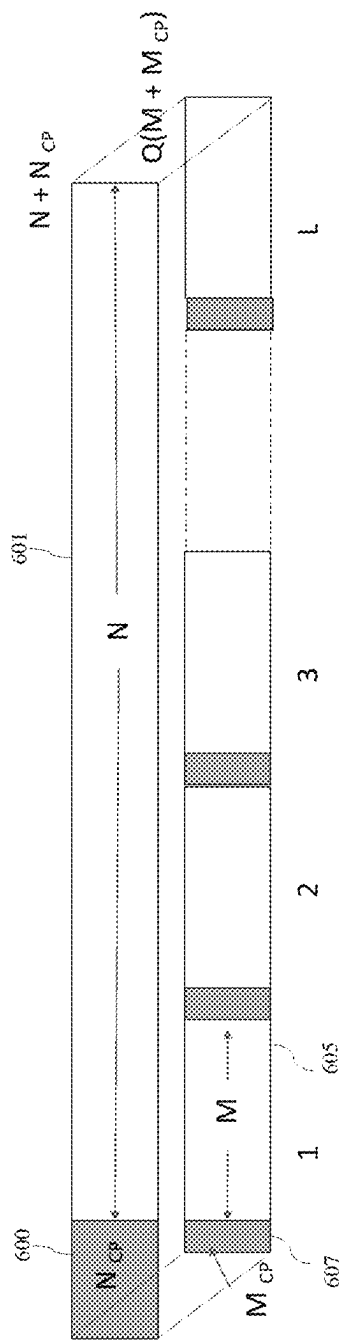


FIG. 6



$$N = M * L$$
$$N_{cp} = M_{cp} * L$$

FIG. 7

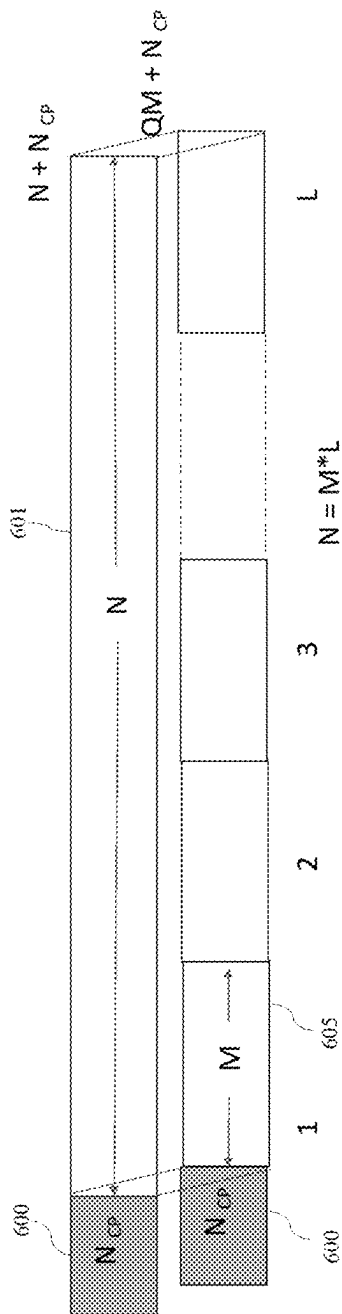


FIG. 8

Service S_0

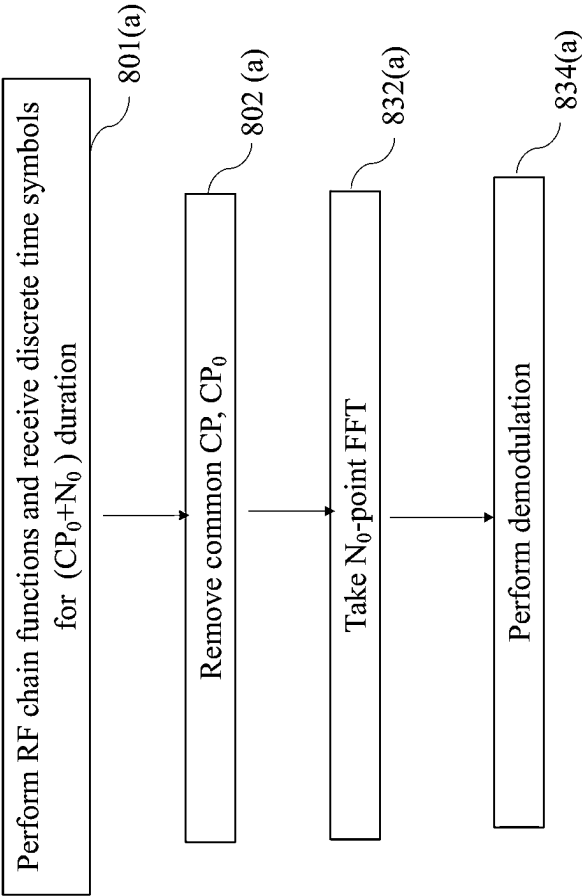


FIG. 9a

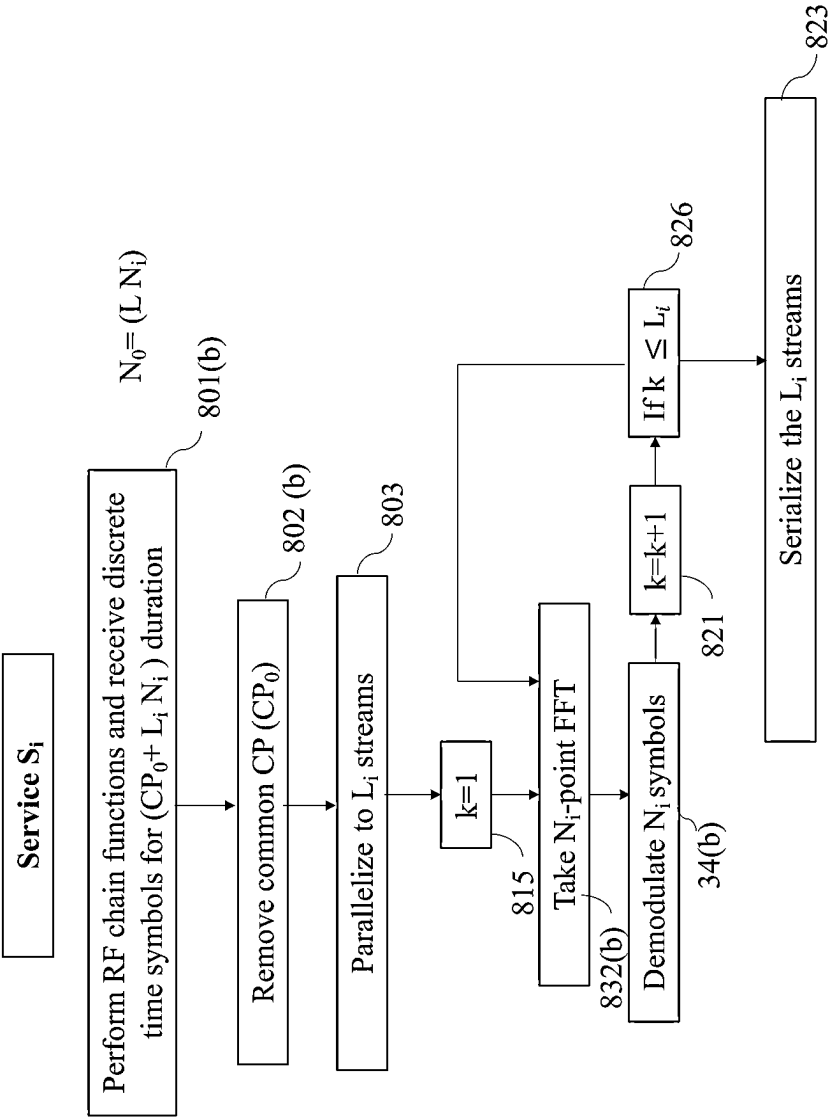
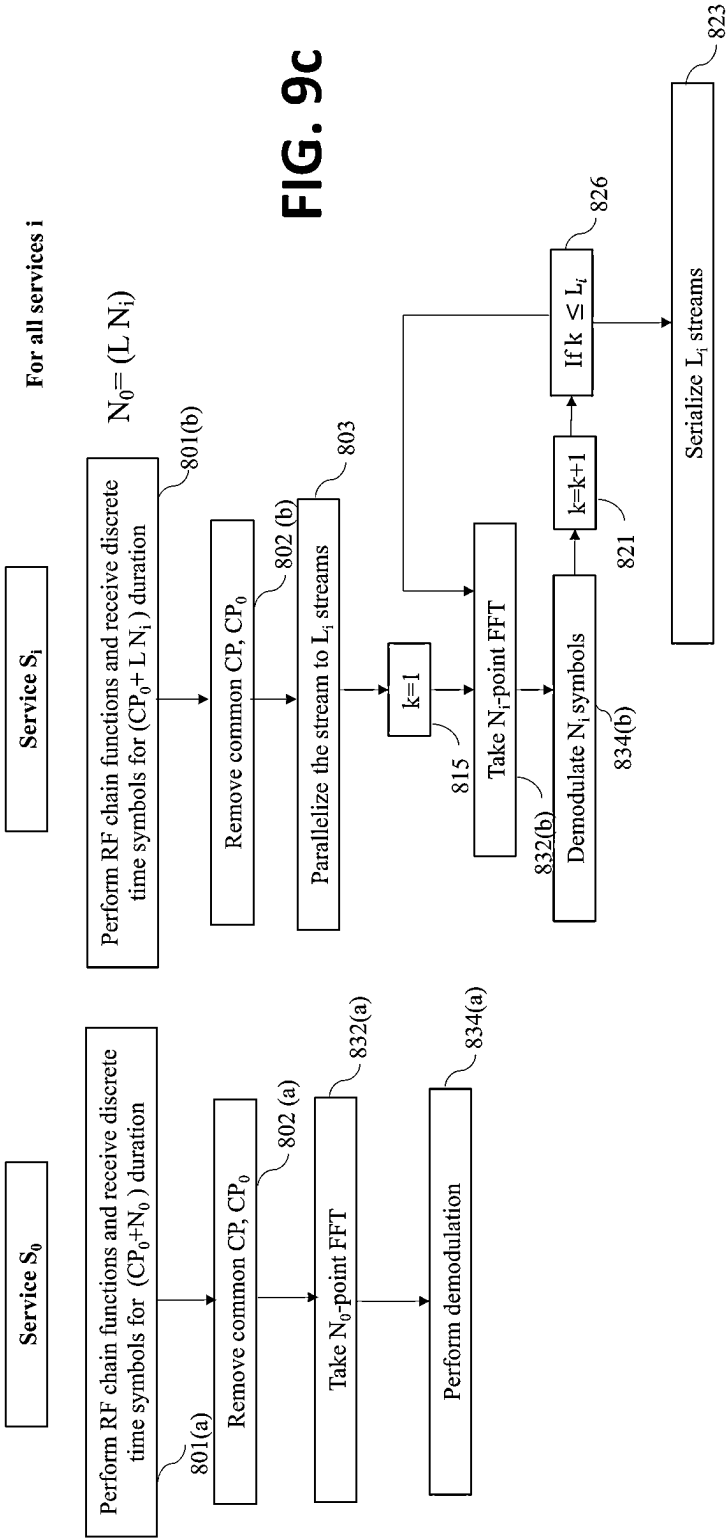


FIG. 9b



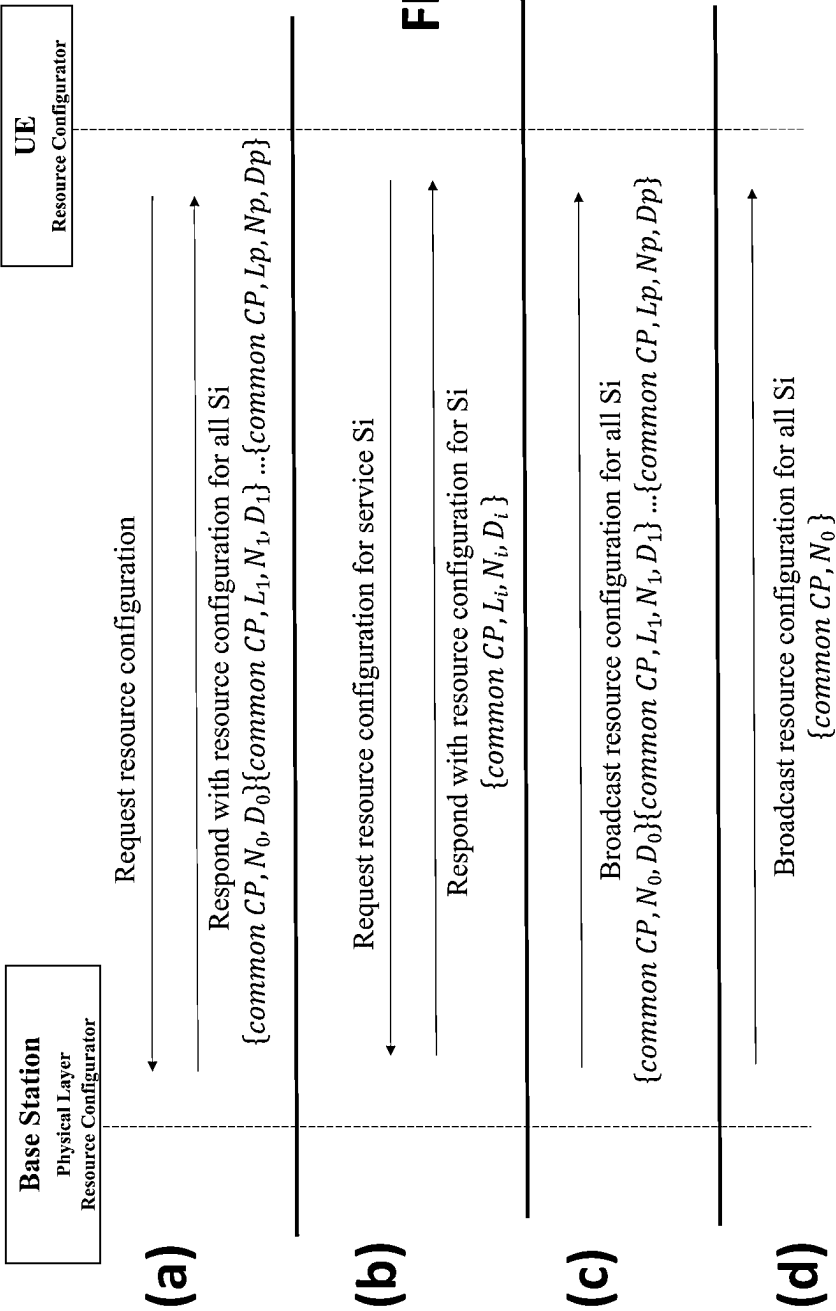


FIG. 10

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METHODS FOR AN IMPROVED RECEIVER FOR MULTI-SERVICE RADIO NETWORKS

CROSS REFERENCE TO THE RELATED APPLICATION

This application claims priority to provisional application 63/294,812 filed on Dec. 29, 2021, the entire content of which is incorporated herein by reference.

TECHNICAL FIELD

The present invention relates generally to wireless networks and more particularly to methods for a receiver within a user equipment supporting multiple service types with Multi-Input and Multi-Output (MIMO) transmitters and beamforming.

BACKGROUND

Any discussion of the prior art throughout the specification should in no way be considered as an admission that such prior art is widely known or forms part of common general knowledge in the field.

The fifth generation (5G) of wireless communication defined by International Telecommunication Union (ITU) comes with three key services to support a wide array of emerging applications. These services are enhanced mobile broadband (eMBB), ultra-reliable low latency communications (uRLLC) and massive machine type communications (mMTC). Each service comes with different radio network performance requirements. For example, eMBB enables an application like high-definition video streaming with up to 10 to 20 Gbps data rates. mMTC leverages benefits of low cost of Internet of Things (IoT) and supports millions of small devices, but the corresponding data rates are in the range of 1 to 100 Kbps. The autonomous vehicle network, an emerging key application, is a key use case for uRLLC that enables highly responsive connections with less than 1 millisecond radio channel latency and high network availability. These services come with conflicting requirements. Deploying them in separate standalone radio systems may seem logical, but it would be highly expensive and inefficient. On the other hand, it is challenging to design a unified radio access network (RAN) which meets the requirements for all types of services without causing interference between services. The interference is particularly important in the downlink direction from the base station (BS) towards the plurality of user equipment (UE) served by that base station wherein all downlink traffic is multiplexed onto a single broadcast (shared) radio channel.

Different service requirements can only be supported by different frequency and time domain implementations. For example, mMTC may require a larger symbol duration to support delay-tolerant devices while uRLLC that has a stringent reliability and latency requirements requires a symbol duration that is very small. One viable solution is to multiplex multiple types of services in one baseband system in orthogonal time and/or frequency resources to avoid the interference.

Orthogonal Frequency division multiplexing (OFDM) is chosen by 5G standards for multiplexing different services due to ease of supporting services with different latency and bandwidth requirements. OFDM is in fact a legacy technique that is known for a long time and has found applications in different types of networks. It simply divides the channel/carrier into a plurality of 'subcarriers' through a

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mathematical function known as Fourier Transform (FT) wherein the subcarriers are waveforms chosen to be 'orthogonal' to each other in frequency domain. Mathematically speaking, when they are orthogonal, the integral of two waveforms over the designated time interval is zero. As a result, orthogonal functions are statistically unrelated.

The concept of using a subcarrier can be visualized as subdividing the total channel into parallel subchannels of smaller bandwidth with frequency-flat response, wherein different waveforms become feasible specific to each subcarrier. The bandlimited orthogonal waveforms are then combined with significant overlap while avoiding inter-subchannel interference. Because subcarriers are orthogonal to each other, they will exhibit no interference, resulting in efficient use of bandwidth. The amplitude of each subcarrier waveform crosses zero at the center of all other subcarriers, minimizing adjacent subcarrier impact.

Due to non-linear characteristics of the radio channel, however, the transmitted time domain signals are unavoidably exposed to what is called 'inter-symbol interference (ISI)'. To remedy ISI, a 'guard interval' (a short time interval) is added between adjacent symbols and a 'cyclic prefix (CP)' is inserted within the guard interval to eliminate ISI coming from the previous symbol. The CP is created in such a way that each OFDM symbol is preceded by a copy of only the end part of the same symbol in time domain. If, for example, the symbol is (1 3 5 7 9 8 8 7 9 1), and CP has length 3, then the prepended symbol becomes (7 9 1 1 3 5 7 9 8 8 7 9 1). CP essentially repeats the end of the symbol so that the linear convolution of a frequency-selective multipath channel can be modeled as circular convolution. When the CP is removed at the receiver (RX), the part of the symbol that is exposed to ISI is essentially removed and the remaining/original symbol is retrieved without any ISI.

FIG. 1 is a basic functional block diagram of a prior art SISO radio system comprised of a single antenna transmitter (TX) 100 (at the base station) and single antenna receiver (RX) 128 (at the user equipment). Let us assume that the transmitter is sending a wideband bit stream to the receiver on the downlink radio channel. We assumed that the transmitter is using N number of subcarriers to transmit. The input bit stream (a series of 0s and 1s) enters the TX on the left side of the diagram. This serial bit stream is first demultiplexed into parallel bit streams by subsystem S/P 101 and then these substreams are fed to the individual Quadrature Amplitude Modulation (QAM) Modulator 102 subsystem, well known in prior art, one for each of the N subcarriers. Each unique grouping of bits is mapped by QAM Modulator into a complex number/symbol (e.g., into a phase and amplitude) in frequency domain. A range of different QAM modulators can be used, starting with BPSK that maps one bit to one symbol, to 256-QAM that maps each 8 bits into one symbol. The series of symbols form the discrete frequency domain representation of the bit stream. Combining modulation with the use of large number of subcarriers results in high throughput.

A key enabler for OFDM is the use of Inverse Fast Fourier Transform (IFFT) 103 subsystem to efficiently create the time domain waveforms (symbols) corresponding to the array of QAM modulated subcarriers shown in FIG. 1. Each subcarrier in time domain simply peaks at the subcarrier frequency. Here, $f_i = i \Delta f + f_0$ and $i=0, 1, \dots, N-1$. Δf is known as the 'subcarrier spacing', N is the number of subcarriers, and f_0 to f_{N-1} defines the frequency channel. While LTE uses a single Δf value that is 15 KHz and a corresponding single CP duration (4.69 μ s), 5G New Radio

(5G-NR) allows different Δf values of 15, 30, 60, 120, and 240 KHz called numerologies of the parent waveform, i.e., 0, 1, 2, 3, and 4 respectively, where the term numerology refers to the different choice of subcarrier spacing, and the resultant OFDM symbol duration and CP duration. In summary, the OFDM subcarriers are $15 \cdot 2^\mu$ KHz, $\mu=(0, 1, 2, 3, 4)$ and the corresponding CP's are $4.69/2^\mu \mu s$ for 5G as shown in TABLE 1. More number of numerologies may be used in future generation networks.

TABLE 1

Parameters	Numerology options (μ)				
	0	1	2	3	4
Subcarrier Spacing (kHz)	15	30	60	120	240
OFDM Symbol Duration (μs)	66.67	33.33	16.67	8.33	4.17
CP Duration (μs)	4.69	2.34	1.17	0.58	0.29
Slot Duration (ms)	1	0.5	0.25	0.125	0.0625

Different numerologies can be used to meet the demands of the services. For example, lower numerologies are more suitable for mMTC, since they can support higher number of simultaneously connected devices within the same bandwidth and require lower power. Intermediate numerologies are appropriate for eMBB which requires both high data rate and sizable bandwidth. Highest numerologies are suitable for delay-sensitive applications pertaining to the uRLLC due to use of shorter symbol duration. Using these different numerologies introduces interference between the different services. With a chosen set of design parameters, the orthogonality no longer holds across different services carried by different numerologies and different number of subcarriers per numerology in the downlink radio channel. The non-orthogonality between services results in inter-numerology interference (INI) known in prior art [5-8] that needs to be addressed.

Going back to FIG. 1, the resulting OFDM signal is first serialized back by subsystem P/S 107 and CP is added to each symbol by subsystem 104. The final signal that is in digital form drives the Digital-to-Analog Converter (D/A) 105 which converts it to an analog radio frequency signal. This baseband signal is usually up-converted to a higher frequency and amplified before being transmitted via the over-the-air channel. At receiver 128, the process is simply reversed. An analog downconverter shifts the OFDM signal back to baseband. The Analog-to-Digital Converter, A/D 106 converts the radio signal back to digital form and passes it to S/P 117 and then to the FFT 113 to transform the discrete time domain signal back to discrete frequency domain signal. The series of QAM Demodulators 112 in turn reproduce the bit stream arriving from each subcarrier, which is then multiplexed by P/S 111 to recreate the original serial data stream. In summary, the key operations are (1) combining a plurality of QAM modulated low-bandwidth subcarriers to create a wideband channel for the user's service, and (2) using FFT and IFFT to efficiently process the symbols in frequency domain. Note that FFT and IFFT processes are simple matrix multiplications well known in signal processing.

With multi-input and multi-output (MIMO) antenna systems of 5G, the radio channel broadcasts the same data stream by each antenna. Different antennas essentially send traffic in different radio path directions using beamforming which enables multiple synchronized transmissions on the

same frequency with minimal or no interference. The RX can retrieve its own data stream from the aggregate data stream.

Precoding (not shown in FIG. 1 for simplicity) is an important process within the TX that is designed based on modeling the current radio channel state information (CSI) in real-time. The goal is to multiply the frequency domain samples of the input at TX with the 'inverse' of the radio channel characteristics to compensate for the channel distortion. In OFDM, an image of so-called 'channel matrix' of each subcarrier is first generated, and then precoding is performed on a per subcarrier basis based on that subchannel realization. At the transmitter side, precoding is performed in frequency domain prior to the IFFT process. The precoder parameters are periodically modified as the radio access network (RAN) performance changes over time. The precoder design is well known in prior art and thus not covered here.

A conventional transmitter supports only a single type of service (as in LTE). Therefore, a single precoder is needed to multiplex data streams of multiple UEs in time domain because all UEs use the same number of OFDM subcarriers, symbol duration and CP duration. Thus, one precoder is designed over each subcarrier based on the corresponding subchannel's frequency. Doing so, the interference between services of different UEs is eliminated. For a multi-service system, however, UEs belonging to different services will use different Fourier transformation (FT) sizes. The FT size N (always a power of 2) is an indicative of the number of subcarriers used for the service. Because each service demands a different bandwidth range, the corresponding FT sizes are different. Furthermore, different services use different numerologies. While service S_0 uses $\mu=0$ ($\Delta f=15$ KHz) with N_0 subcarriers, another service S_1 uses $\mu=4$ ($\Delta f=240$ KHz) with N_1 subcarriers, which means that they will not experience the same frequency-domain radio channel representation. Those services with a larger number of samples (i.e., larger FFT size) will distribute symbols to large number of OFDM subcarriers while services with smaller number of samples (i.e., smaller FFT size) will use only a subset of the available subcarriers. Precoding will be applied to those subcarriers in-use for the service, and not applied to those subcarriers that are not used by the service. As a result, each service requires a separate precoder at the transmitter (e.g., one with precoding at each N_0 subcarrier, and the other one with precoding only at N_1 subcarriers). Note that precoding requires a lot of computation. For an M_T -antenna transmitter and M_R antenna receivers, and N subcarriers, the precoder must perform precoding for $(M_T M_R N)$ subchannels. These subchannel parameters change over time when the radio channel performance changes. In summary, the frequency response of the channel is different from one service to another and from time to time. With different FFT sizes for two services, the direct precoding approach would require two different precoders, ZF_1 and ZF_2 performing precoding for $(M_T M_R N_0)$ and $(M_T M_R N_1)$ subchannels, respectively. The interference from smaller size FFT services to larger size FFT services in a multi-service configuration of the transmitter results in inter numerology interference (INI). The precoding will eliminate INI across those subcarriers commonly used by both services. However, INI from UEs using S_0 will not be eliminated across the rest of the subcarriers that only S_1 is using. Furthermore, using additional precoders means additional transmitter complexity and cost. A high price to pay when INI is only partially eliminated.

In summary, the INI problem is partially addressed in frequency domain in prior art by using a plurality of pre-

coders. The main problem is that ‘unmatched’ number of subcarriers will introduce interference as precoding will not be performed to unseen subcarriers by UEs. More specifically, those UEs with smaller FFT size will interfere on UEs belonging to larger FFT sizes due to uncommon subcarriers. The INI problem can be addressed in time domain to increase spectral efficiency.

A list of prior art related to the instant disclosure:

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SUMMARY

Technical Problem

Based on aforementioned discussions, an object of the present invention devised to solve the conventional INI problem in time domain is to provide a method of using the same number of subcarriers and the same (common) CP across all services for the multi-service broadcast radio channel, and an apparatus that implements the method

within a MIMO base station transmitter without introducing additional componentry to the receiver in the UE.

Technical Solution

According to an aspect of this invention, the complexity is reduced, and at the same time, INI is eliminated. There is a single ZF precoder regardless of the number of services. The innovative concept is to perform ZF precoding across all possible subcarriers regardless of whether they are being used by a service or not. To enable this simple concept, a “frequency resampling” operation is performed before precoding to equalize the number of subcarrier usage across all services. The FFT size of the service, S_0 , that uses the largest number of subcarriers (and hence the longest OFDM symbol) is taken as the ‘reference’ FFT size, N_0 . The ‘reference’ can be, by default, numerology zero or the lowest numerology of active services. The frequency resampling operation is performed for the i^{th} service, S_i , in its Frequency Re-sampler, V_i . First, V_i performs L_i times N_i IFFT operation for a given S_i , that uses N_i subcarriers of numerology say μ -corresponding symbol duration is D_i and CP duration is CP_i (see TABLE 1)—and obtain the corresponding time domain samples. Then, these time domain samples are converted back to the frequency domain, but with a ‘common’ FFT size of N_0 (indicative of common number of subcarrier usage). This in effect results in multiple concatenated L_i OFDM symbols at a subcarrier to become one large OFDM symbol prior to ZF precoding. Note that no frequency resampling is needed for the reference service. All other services with smaller FFT sizes are converted into the reference FFT size before precoding. In this case, regardless of the number of different types of services, only $(M_T M_R N_0)$ number of channel precoding is needed as the number of subcarriers is equalized across all services.

As an example, let the reference service be denoted by S_0 and have an FFT size of N_0 . The i^{th} service, S_i has an FFT size of $N_i = N_0 / L_i$, where L_i is an integer. Note that the FFT size is a direct indicative of the numerology selected by the service. A higher numerology corresponds to lower the FFT size. In S_i , the number of samples are taken back to the frequency domain for $N_0 = L_i * N_i$ samples. The precoder is designed over N_0 subcarriers for all services. After precoding, an IFFT operation with size N_0 is performed as if one OFDM symbol of this size is to be transmitted. Thus, the added CP is the same for all services and its size is equal to that of the largest CP service, i.e., CP_0 . Only service S_0 will have one CP (CP_0) per symbol. All other services, S_i , will have CP_0 (common CP) encapsulating L_i number of symbols that correspond to S_i ’s numerology.

At the receiver of the UE, the common CP, CP_0 , of service S_i is removed every L_i OFDM symbols. This is the only difference at the receiver side compared to conventional OFDM receivers. Therefore, in general, the complexity is reduced in the proposed scheme compared with the prior art where each service has its own precoder.

The implementation according to this invention requires the receiver to know about the common CP usage, i.e., CP_0 removal every L_i symbol for service type S_i . This can be achieved according to various exemplary embodiments listed below:

In one embodiment, the numerology zero service can be pre-programmed at each receiver for the common CP usage. However, a pre-programmed fixed UE resource configuration does not give sufficient flexibility to accommodate the following exemplary cases:

The lowest numerology service in the network may not necessarily be zero.

A legacy base station that the UE receives service from may only support variable size CP in accordance with prior art.

The transmitter may operate in a dual-mode, i.e., may want to switch from common CP to variable CP and vice versa.

To remedy the above shortcomings, the use of common CP along with the corresponding reference service numerology can be instructed to all receivers by the transmitter using a control message. This method uses a broadcast control connection/channel (BCC) between the base station and all connected UEs to send the control message.

In one embodiment, the configuration parameters for all service types may be sent in the control message.

In another embodiment, the configuration parameters for all service types may be readily preprogrammed on the receiver. Once the base station sends the common CP usage instruction with a choice of reference service numerology, the receiver can use this information for determining the proper configurations.

Since all UEs are configured for all service types, this method has a shortcoming of configuring resources for services that the UE may not be using.

The receiver may send a control message to the base station requesting resource configuration parameters for only specific services or for one service type. The transmitter sends back the reference service numerology to use in common CP. This method uses a dedicated control connection (DCC) between the base station and UE for sending and receiving control messages. The transmitter in the base station may receive a 'request' from the receiver in the form of a control message, and in turn, the transmitter 'responds' in the form of a control message by providing the requested resource configurations.

In one embodiment, the transmitter in the base station may autonomously send resource configuration information for a plurality of service types or a single service type without needing a 'request' from any of the UEs. This embodiment uses a broadcast control connection between the transmitter and all UE receivers because the same resource configuration information goes to all UEs.

In one embodiment, the resource configuration information that the transmitter sends to the receiver includes data such as the subcarrier spacing (numerology), number of subcarriers to use, N_s , the variable duration cyclic prefix CP_v , or common duration cyclic prefix CP_o , the number of OFDM symbols in a frame between successive CPs, L_f . In some embodiment, the transmitter may decide to alter from time-to-time said resource configuration information.

In one embodiment, the transmitter only communicates in a control message the resource configuration information of 'common CP' and 'reference service type' without communicating any other resource information wherein said any other resource information is pre-programmed and known within the receiver for each service type. In this embodiment, the receiver determines common CP duration, CP_o (based on CP of the specified reference service type) and L_f ($=N_o/N_s$, wherein N_s and N_o are pre-programmed) for each service type S_i .

In one embodiment, unless otherwise specified by the base station, the receiver may use the prior art variable CP duration method by default. In another embodiment, unless otherwise specified by the base station, the receiver may use the common CP duration method with numerology zero by default.

In one embodiment, the reference service type may always be the numerology zero service. In another embodiment, the reference service type is the lowest numerology (largest FFT size) service.

In one embodiment, the transmitter may provide the first resource configuration data corresponding specifically to ultra-reliable and low-latency (uRLL), second resource configuration data associated with enhanced mobile broadband (eMBB), and third resource configuration data associated with mMTC service.

In one embodiment, the base station is capable of transmitting said resource configuration data corresponding to all service types in one control message. In another embodiment, the base station is capable of transmitting said resource configuration data corresponding to each service type in a separate control message.

In some embodiments, the base station receives from a user equipment (UE) the resource configuration request control message using a pre-configured frequency sub-band allocated only for control messaging during synchronization between base station and UE, and a first subcarrier spacing, and transmitting a resource configuration response message using said sub-band and first subcarrier spacing, or another pre-configured sub-band, and a second subcarrier spacing.

The resource configuration control messaging uses the radio channel between antenna pairs just like the data traffic, and therefore the modulation and demodulation steps described herein applies to control messages as well.

All such flexibilities (and others not mentioned here) on configuration of the receiver for the use of common or variable CP based on the number of BS antennas, receiver capabilities, and service mix are trivial extensions and thus covered by this invention.

Advantageous Effects

According to the embodiment of the present invention, the UE receiver does not require additional hardware components, or additional complexity compared to prior art 5G UE receiver. It only needs (i) a method for receiving resource configuration control messages on common CP usage, and (ii) another method to remove common CP every L_f symbol for service S_i , as opposed to removing common CP every symbol for the reference service, S_o .

It will be appreciated by persons skilled in art that the effects that could be achieved with present invention are not limited to what has been particularly described here. Other advantages can be clearly understood from the following detailed description.

BRIEF DESCRIPTION OF THE DRAWINGS

The present disclosure, in accordance with one or more various examples, is described in detail with reference to the following figures. The drawings are provided for purposes of illustration only and merely depict examples of the disclosure. These drawings are provided to facilitate the reader's understanding of the disclosure and should not be considered limiting of the breadth, scope, or applicability of the disclosure. It should be noted that for clarity and ease of illustration these drawings are not necessarily made to scale.

FIG. 1 is a diagram illustrating a simple prior art configuration of TX and RX.

FIG. 2 is a diagram illustrating key modulation functions of a MIMO TX for a single service.

FIG. 3 is a diagram illustrating simple prior art configuration for removing INI.

FIG. 4 is a diagram illustrating the modulation configuration of TX for removing INI according to invention.

FIG. 5 is a diagram illustrating the configuration of RX according to invention

FIG. 6 is a diagram illustrating the Frequency Re-sampler according to invention.

FIG. 7 is a diagram illustrating the prior art CP for two services.

FIG. 8 is a diagram illustrating common CP for two services according to invention.

FIG. 9a is a diagram illustrating first method of the receiver according to invention.

FIG. 9b is a diagram illustrating second method of the receiver according to invention.

FIG. 9c is a diagram illustrating third method of the receiver according to invention.

FIG. 10 is a diagram illustrating various control messaging options according to invention.

Below is a list of acronyms used in the instant application:

A/D Analog to Digital
 3GPP 3rd Generation Partnership Program
 BCC Broadcast Control Channel
 BER Bit Error Rate
 BPSK Binary Phase Shift Keying
 BS Base Station
 CP Cyclic Prefix
 CSI Channel State Information
 D/A Digital to Analog
 DCC Dedicated Control Channel
 DL Down Link
 eMBB Enhanced Mobile Broadband
 FFT Fast Fourier Transform
 IFFT Inverse Fast Fourier Transform
 INI Inter Numerology Interference
 IoT Internet of Thing
 ISI Inter Symbol Interference
 MIMO Multi Input Multi Output
 mMTC Massive Machine Type Communications
 OFDM Orthogonal Frequency Division Multiplexing
 QAM Quadrature Amplitude Modulation
 RAN Radio Access Network
 RF Radio Frequency
 RX Receiver
 SISO Single Input Single Output
 SNR Signal to Noise Ratio
 TX Transmitter
 UE User Equipment
 uRLLC Ultra-Reliable Low-Latency Communications
 ZF Zero Forcing

DETAILED DESCRIPTION OF THE EMBODIMENTS

The first aspect of invention is a method executed in a receiver of a user equipment (UE) in a wireless OFDM radio access network (RAN) receiving data over the broadcast radio channel from a base station through M_R receive antennas with possible configurations of MIMO, SISO, SIMO or MISO, for first service type using a lower numerology and second service type using a higher numerology, wherein resource configuration for first service type selected as the reference service by the base station and communicated by said base station to UE as using N_0 number of subcarriers, subcarrier spacing of Δf_0 , and cyclic prefix duration of CP_0 and resource configuration for second service having been defined as using N_1 number of subcarriers, subcarrier spacing of Δf_1 and cyclic prefix duration of CP_1 ,

wherein $L=N_0/N_1$ and $CP_1=CP_0/(N_0/N_1)$ and L is an integer that is positive powers of 2, said receiver applying the steps of: (a) Removing the CP of duration CP_0 from each time domain symbol generated by first service, (b) Removing the CP of duration CP_0 from each L number of consecutive time domain symbols generated by second service, (c) Taking N_0 point FFT of time domain symbols of first service obtained in step (a) to create corresponding N_0 frequency domain symbols, (d) Taking L times N_1 point FFT of time domain symbols of second service obtained in step (b) to generate corresponding LN_1 frequency domain symbols, (e) Demodulating N_0 frequency domain symbols generated in step (c) into N_0 bit streams, (f) Demodulating LN_1 frequency domain symbols generated in step (d) into LN_1 bit streams, (g) Multiplexing bit streams of steps (e) into a serial stream of bits corresponding to first service, (1) Multiplexing bit streams of steps (f) into a serial stream of bits corresponding to second service. Here lower numerology can be numerology zero.

The second aspect of invention is a method executed in a receiver of a user equipment (UE) in a wireless OFDM radio access network receiving data over the broadcast radio channel from a base station through M_R receive antennas with possible configurations of MIMO, SISO, SIMO or MISO, for first service that is selected as the reference service by the base station and communicated by said base station to UE, is using the lowest numerology and a plurality of other service types using higher numerology than reference service type, resource configuration for first service type (reference service) having been communicated by said base station to UE as using N_0 number of subcarriers, subcarrier spacing of Δf_0 , and cyclic prefix duration of CP_0 and resource configuration for said plurality of other service types been defined as, for service type S_i , using N_i number of subcarriers, subcarrier spacing of Δf_i and cyclic prefix duration of CP_i , wherein $L_i=N_0/N_i$ and $CP_i=CP_0/(N_0/N_i)$ and L_i is an integer that is positive powers of 2, said receiver applying the steps of: (a) Removing the CP of duration CP_0 from each time domain symbol generated by first service, (b) Removing the CP of duration CP_0 from each L_i number of consecutive time domain symbols generated by second service, (c) Taking N_0 point FFT of time domain symbols of first service obtained in step (a) to create corresponding N_0 frequency domain symbols, (d) Taking L_i times N_i point FFT of time domain symbols of second service obtained in step (b) to generate corresponding $L_i N_i$ frequency domain symbols, (e) Demodulating N_0 frequency domain symbols generated in step (c) into N_0 bit streams, (f) Demodulating $L_i N_i$ frequency domain symbols generated in step (d) into $L_i N_i$ bit streams, (g) Multiplexing bit streams of steps (e) into a serial stream of bits corresponding to first service, (1) Multiplexing bit streams of steps (f) into a serial stream of bits corresponding to service S_i . Here lower numerology can be numerology zero.

The third aspect of invention is a physical layer resource configuration method executed in a receiver of a UE in an OFDM wireless radio access network receiving data on broadcast radio channel from a base station for a plurality service types, each service using a different numerology, said configuration method has the steps of: (a) User Equipment sending a control message to said base station requesting physical layer resource configuration to use for each service type, (b) User Equipment receiving from said base station a control message in response to its requesting that includes resource configuration that specifies common cyclic prefix duration of CP_0 , and other resource configuration information for each service type wherein other

resource configuration information for each service type S_i are N_i number of subcarriers, and subcarrier spacing of Δf_i .

The fourth aspect of invention is a physical layer resource configuration method executed in a receiver of a UE in an OFDM wireless radio access network receiving data on broadcast radio channel from a base station for a plurality of service types, each service using a different numerology, said configuration method has the step of User Equipment receiving from time to time a control message from said base station that includes common cyclic prefix duration of CP_0 and other resource configuration for each service type wherein other resource configuration information for each service type S_i are N_i number of subcarriers, and subcarrier spacing of Δf_i .

While this invention is illustrated and described in a preferred embodiment, the invention may be produced in many different configurations. There is depicted in the drawings, and will herein be described in detail, a preferred embodiment of the invention, with the understanding that the present disclosure is to be considered as an exemplification of the principles of the invention and the associated functional specifications for its construction and is not intended to limit the invention to the embodiment illustrated. Those skilled in the art will envision many other possible variations within the scope of the present invention.

Note that in this description, references to “one embodiment” or “an embodiment” mean that the feature being referred to is included in at least one embodiment of the invention. Further, separate references to “one embodiment” in this description do not necessarily refer to the same embodiment; however, neither are such embodiments mutually exclusive, unless so stated and except as will be readily apparent to those of ordinary skill in the art. Thus, the present invention can include any variety of combinations and/or integrations of the embodiments described herein.

Radio Access Network (RAN) has many more components than those described in this disclosure. Those components are omitted so that the invention is easy to understand. Furthermore, other parts of a new generation wireless networks such as the data plane and control plane of the core network are also omitted as those parts are not relevant to invention.

MIMO is the acronym for Multiple Input Multiple Output. This means that the radio is split up in multiple identical radio frequency (RF) chains that can all transmit and receive data individually and simultaneously with a plurality of antennas. More data can, therefore, be sent or received simultaneously, or alternatively the same data can be sent or received several times for reliable transmission. A radio frequency (RF) chain may include a broad range of discrete components such as attenuators, switches, amplifiers, detectors, synthesizers, and other analog components along with A/D and D/A converters. In what follows, subcomponents specific to RF chain that are well known in prior art are also omitted to keep the descriptions focused on invented methods and subcomponents. Furthermore, at higher frequencies the sizes of antennas decrease where many antennas might be compacted in a small volume. The increase in the number of elements makes it costly to have a dedicated RF chain for every single antenna. Therefore, the number of RF chains might be less than that of antenna elements. This feature is enabled by an RF precoder which is well known in prior art. However, for the sake of simplicity, the number of antennas is assumed to be the same as the number of RF chains, M_T since RF precoder is out of scope. The number of RF chains and antennas are thus used interchangeably.

Furthermore, functionalities according to several embodiments of this invention are grouped as transmitter functions of a base stations for simplicity, wherein in another embodiment of this invention some functionalities may belong to subsystems of the base station other than the transmitter.

In some embodiment, the subcarrier spacing in OFDM refers to the frequency interval between allocated subcarriers. Through the subcarrier spacing, OFDM enables data transmission while maintaining orthogonality among subcarriers. LTE supports only one subcarrier spacing that is fixed at 15 kHz. The ‘numerology’ refers to how to configure a unit of radio access network resource in time domain. LTE RAN uses a single numerology with 15 kHz as the subcarrier spacing, while OFDM RAN uses various numerologies each one supporting a different subcarrier spacing (an integer multiple of 15 KHz) to support a wide frequency range up to 100 GHz and hence wider bandwidth than LTE, and various service types eMBB, uRLL, and mMTC with varying requirements.

In some embodiments, an OFDM symbol duration comprises a cyclic prefix (CP) duration and a duration for the symbol in time-domain in a unit of microseconds. The transmitted OFDM symbol duration is determined as the sum of the two durations for the CP and the output of IFFT.

In some embodiments, a UE is configured to receive the two services but configured with the same set of numerology parameters for these two services. In some embodiments, a UE is configured to receive the two services but configured with two different sets of numerology parameters for these two services. In such embodiments, the numerology parameters include the CP length, the subcarrier spacing, OFDM symbol length, FFT size (indicative of the number of subcarriers used).

FIG. 2 illustrates the block diagram of a prior art transmitter of an OFDM MIMO base station with M_T number of RF chains (assuming equal to number of antennas) and a single type of service (assuming at numerology 0: subcarrier spacing is 15 KHz). The input data is a series of bits that arrive at the base station from the core IP network. These bits are first converted to frequency domain symbols using a QAM modulator as depicted in FIG. 1. The modulators and S/P multiplexers are omitted from the diagram for simplicity.

The input to ZF Precoding **201** is a series of symbols parallelized on a per subcarrier basis wherein subcarrier spacing is 15 KHz. ZF Precoding **201** multiplies each arriving symbol with the inverse of the that subcarrier’s channel state information (CSI) and sends these processed symbols as inputs to N-point inverse FFT **202** (we assumed the number of subcarriers being equal to FFT size N). Block **203** adds the same duration CP to each time domain symbol. After multiplexing of all symbols from all subcarriers, Block **204** performs all RF Chain functions to translate discrete time symbols to actual analog radio signals and transmits them using antenna **205**.

In a MIMO base station, above operations are performed across all RF chains. This parallelization is needed as ZF precoding may see different channel characteristics as reflected by each base station RF chain (1, 2, . . . , M_T) and each UE RF chain (1, 2, . . . , M_R) that are causing radio signals in spatially different directions. Therefore, there are ($M_T M_R$) distinct radio paths to consider. Furthermore, ZF Precoding performs precoding for each subcarrier resulting in a total of $M_T M_R N$ computations for N subcarriers per input stream of symbols.

FIG. 3 illustrates the block diagram of a prior art transmitter of an OFDM MIMO base station with M_T number of

antennas and two types of service with two numerologies and corresponding different FFT sizes, N_1 and N_2 , and different CP sizes, CP_1 and CP_2 . The input data is a series of bits that arrive at the base station from the core IP network. These bits are first converted to frequency domain symbols using a QAM modulator as depicted in FIG. 1. The modulators and S/P multiplexers are omitted from the diagram for simplicity.

The input to ZF Precoding **301** is a series of symbols parallelized on a per subcarrier basis according to the first service's subcarrier spacing. ZF Precoding **301** multiplies each arriving symbol with the inverse of the that subcarrier's channel state information (CSI) and sends these processed symbols as inputs to N_1 -point inverse FFT **302**. Block **303** adds CP_1 to each time domain symbol. All symbols from all subcarriers are then multiplexed.

A parallel process is executed by ZF Precoding **311** to the second service. ZF Precoding **311** multiplies each arriving symbol with the inverse of the that subcarrier's channel state information (CSI) and sends these processed symbols as inputs to N_2 -point inverse FFT **312** (we assumed the number of subcarriers being equal to FFT size). Block **313** adds CP_2 to each time domain symbol. After multiplexing of all symbols from all subcarriers. Block **304** performs the RF Chain functions to the merged symbols and translate discrete time symbols to analog radio signals. Analog signals are sent to antenna **305**.

The total number of ZF precoding operations according to prior art configuration of FIG. 3 when there is a reference service S_0 with N_0 subcarriers, and other p number of services of different numerology, each service of different numerology, S_i , having N_i subcarriers, and $L_i = N_0/N_i$ is:

$$M_T M_R N_0 + \sum_{i=1}^p (M_T M_R (L_i N_i)) \quad (\text{Eq. 1})$$

The total number of IFFT operations according to said prior art configuration is:

$$M_T N_0 \log_2(N_0) + M_T \sum_{i=1}^p (L_i N_i \log_2(N_i)) \quad (\text{Eq. 2})$$

wherein $N_0 \log_2(N_0)$ is the number of IFFT operations are performed for service S_0 per antenna and $L_i N_i \log_2(N_i)$ is the number of IFFT operations per each service S_i per antenna. The formula for number of N point-FFT operations is well-known in prior art.

FIG. 4 illustrates the block diagram of the OFDM MIMO base station with transmitter **451**, M_T number of antennas **425**, Physical Layer Resource Configurator **441**, and a plurality of services, $S_0, S_1, \dots, S_p$ according to invention. The UE is represented by common CP MIMO OFDM receiver **452** with M_R antennas **435**, and UE Resource Configurator **442**. The services use different numerologies and different FFT sizes, $N_1, N_2, \dots, N_p$ and a common CP size, CP_0 , that correspond to reference service S_0 . The bit streams of each service (except reference service) first traverse Frequency Re-sampler **413**, before ZF precoding. After frequency resampling all streams are essentially redistributed to N_0 subcarriers. Thereafter, N_0 -point FFT **422** and CP_0 Adder **423**, and RF Chain **424** process the symbols just as in FIG. 3. The only difference is that now all services are mapped to equal number of subcarriers.

FIG. 4 illustrates Physical Layer Resource Configurator **441** subsystem which may be a subcomponent of the transmitter. Alternatively, it may be a subcomponent of another component of the base station. It communicates with UE Resource Configurator **442** to set the transmit parameters such that UE is informed on usage of common CP on downlink channel. The messaging is performed on radio link **443** which may be a dedicated subchannel for control

messaging or a subchannel shared with other control messages between the BS and UE.

In one embodiment, the transmitter in the base station operates only according to common CP of this invention as illustrated in FIG. 4 (single mode). In another embodiment, the transmitter in the base station is 'dual-mode', i.e., accommodates both common CP method and the prior art method of variable CP that is illustrated in FIG. 3. Dual-mode transmitter can switch from common CP mode to variable CP mode, or vice versa, or it can be pre-configured according to either common CP mode or variable CP mode. The dual-mode is beneficial when the UEs don't support common CP, or when the base station has some operational reasons for preferring variable CP.

One method of dual-mode transmitter using variable CP is activating componentry such as the additional ZF precoders and per-service IFFT operations (shown in FIG. 3) while deactivating Frequency Re-samplers (shown in FIG. 4). Another method of dual-mode transmitter using common CP is deactivating componentry such as the additional ZF precoders and per-service IFFT operations while activating Frequency Re-samplers.

FIG. 5 illustrates the block diagram of receiver **452** at the UE. For services S_0 and S_i , UE choses to use the RF streams coming from antenna **513** amongst the plurality of antennas because the received radio signal has the highest power. RF Chain **515** converts received analog symbols to discrete symbols. CP_0 Remover **516(a)** removes the common CP from every symbol for service S_0 . Subsequently, N_0 -point FFT **517(a)** converts time domain symbols to frequency domain symbols. Finally, QAM Demodulator **527(a)** converts each symbol into a series of bits. On the other hand, CP_0 Remover **516(b)** removes the common CP from every L_i symbol for service S_i . Subsequently, N_i -point FFT **517(b)** converts these L_i symbols from time domain to frequency domain symbols. Finally, QAM Demodulator **527(b)** converts each symbol into a series of bits. Aforementioned functions for each service type can be implemented by the same components, i.e., CP Remover **517(a)** and **(b)** may be the same component, or different parallel components. Different hardware and software implementation of these functions are possible and covered by this invention.

FIG. 6 illustrates a simple block diagram of Frequency Re-sampler **413** shown in FIG. 4 the function of which is to translate a bit stream modulated to $L_i N_i$ frequency domain symbols to N_0 frequency domain symbols spread across N_0 subcarriers. The embodiment is for illustration of functionality only. After parallelizing the bit stream N_i bit streams, QAM_1 to QAM_{N_i} **515** translates them into N_i symbol stream and feed as input to Frequency Resample **413**. Subsequently, N_i -point IFFT **516** subsystem converts the modulated symbols into N_i time domain symbols. This process is repeated L_i times. P/S **517** multiplexes the symbols into a serial symbol stream which is then demultiplexed to N_0 streams by S/P **518**. N_0 -point FFT **519** subsystem takes these N_0 time domain symbols as input and translates them back to N_0 frequency domain symbols to be spread to N_0 subcarriers.

The total number of ZF precoding operations according to embodiments of this invention when there is a reference service S_0 with N_0 subcarriers, and other p number of services of different numerology, each service of different numerology having N_i subcarriers, and $L_i = N_0/N_i$ is:

$$M_T M_R N_0 \quad (\text{Eq. 3})$$

The corresponding total number of IFFT and FFT operations (including the Frequency Re-sampler) according to invention is:

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$$M_T N_0 \log_2(N_0) + (\sum_{i=1}^P (L_i N_i \log_2(N_i) + N_0 \log_2(N_0))) \quad (\text{Eq. 4})$$

wherein the first term of $M_T N_0 \log_2(N_0)$ is the number of IFFT operations that are performed for M_T antennas. The second term has $(L_i N_i \log_2(N_i) + N_0 \log_2(N_0))$ that is the number of back-to-back IFFT and FFT operations, respectively, performed by each Frequency Re-sampler V_i .

Just to give a simple measure of complexity reduction with the present invention Eqs. 1 and 3 are compared, and the number of precoding savings with the invention are determined to be $\sum_{i=1}^P (M_T M_R (L_i N_i))$. Similarly, comparing Eqs. 2 and 4, the number of FFT/IFFT operations savings with the invention is $(M_T \sum_{i=1}^P (L_i N_i \log_2(N_i)) - \sum_{i=1}^P (L_i N_i \log_2(N_i) + N_0 \log_2(N_0)))$ wherein the first term here dominates with large number of antennas in MIMO transmitter configurations. Other savings are due to reduced number of precoders and INI elimination operations.

FIG. 7 illustrates example symbol structures in the time domain for a RAN to support two different services according to prior art wherein the two services use different numerologies. An embodiment of the frame structure for a network to support two services shown in FIG. 7 is for illustration only. Service 1 uses a symbol length of N with a CP length of N_{cp} . Service 2 uses a symbol length of M with a CP length of M_{cp} . These lengths are durations and pre-defined by the numerology selected for the service type associated with Service 1 and 2. Note that N_{cp}/M_{cp} is the same as N/M and represented by an integer number L (note that N and M are integer powers of 2). Stated otherwise, when the symbol length decreases the corresponding CP decreases proportionally. The UE receiving Service 1 and Service 2 symbols removes the corresponding CPs and retrieve the actual symbols of lengths N and M , respectively. Note that Service 1 and Service 2 uses different subcarrier spacing and different number of subcarriers according to their numerology. For example, Service 1 may use numerology 0 with subcarrier spacing of 15 KHz and FFT size of 4096, and Service 2 may use numerology 1 with subcarrier spacing 30 KHz and FFT size of 2048.

FIG. 8 illustrates example symbol structures in the time domain for a RAN to support two different services according to invention wherein the two services use different numerologies. An embodiment of the frame structure for a network to support two services shown in FIG. 8 is for illustration only. Service 1 uses a symbol length of N with a CP length of N_{cp} . The common CP length, CP_0 is N_{cp} , assuming that Service 1 is the reference service S_0 . Service 2 uses a symbol length of M where $N/M=L$ and L is an integer. These lengths are pre-defined by the numerology selected for the service type associated with Service 1 and 2. Note that these M symbols use only one CP that is the common CP. Stated otherwise, the symbol length decreases from N to M , but the corresponding CP that covers M symbols stay as constant length. The UE receiving Service 1 removes the corresponding CP of length N_{cp} for each symbol and retrieve the actual symbol of length N . The UE receiving Service 2 however removes the corresponding CP of length N_{cp} for every M symbol and retrieve the actual symbol of length M for L consecutive symbols thereafter. Note that Service 1 and Service 2 uses different subcarrier spacing and different number of subcarriers with their respective subcarrier spacing. For example, Service 1 may use numerology 0 with subcarrier spacing of 15 KHz and 4096 subcarriers, and Service 2 may use numerology 1 with subcarrier spacing 30 KHz and 2048 subcarriers.

FIG. 9a shows a simple flow chart depicting the key steps of handling an incoming radio stream at the receiver for the

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reference service according to prior art that uses only numerology zero for all services (e.g., for LTE). At step 801(a), analog radio signals for $(CP_0 + N_0)$ duration are received and the RF chain functions are performed to retrieve corresponding discrete time symbols. In step 802(a), the common CP, CP_0 is removed. Then in step 832(a), the receiver takes N_0 -point FFT to convert these symbols back to frequency domain. Step 834(a) performs the demodulation to retrieve the corresponding bit stream.

FIG. 9b shows a simple flow chart depicting the steps of handling an incoming radio stream at the receiver for a single non-reference service according to usage of common CP. This scenario is applicable to a UE that is an IoT (mMTC service) or an autonomous car (uRLLC service) that is using only one service type but of a different numerology than the reference service. At step 801(b), analog radio signals for $(CP_0 + L_i N_i)$ duration are received and the RF chain functions are performed to retrieve corresponding discrete time symbols. In step 802(b), the common CP, CP_0 is removed.

In one embodiment, the remaining symbols are parallelized to L_i streams first in step 803. Then, in step 832(b), the receiver takes N_i -point FFT to convert the symbols back to frequency domain. Step 834(b) performs the demodulation to retrieve the bit stream corresponding to each symbol. This processing loop repeats L_i number of times according to steps 821(b) and 826(b) until all symbols are demodulated. In step 823, these L_i streams are serialized into a bit stream.

In another embodiment, the parallelization step of 803 is not performed. Instead, the first N_i symbols enter N_i -point FFT according to step 832(b) that converts the symbols back to frequency domain and subsequently demodulation according to step 834(b) retrieves the bit stream corresponding to each symbol, and then process continues for the second batch of N_i symbols, and so on until all $L_i N_i$ symbols are processed.

FIG. 9c shows a simple flow chart of the most generalized case of depicting the steps of handling an incoming radio stream in the receiver that has a mix of reference service and a plurality of non-reference services according to usage of common CP. Two distinct paths are shown: First path applies to the reference service S_0 (same steps shown in FIG. 9a) and the second path applies to all non-reference services S_i (same steps shown in FIG. 9b).

FIG. 10 shows various embodiments of control messaging between Physical Layer Resource Configurator 441 located in the base station and Resource Configurator 442 located in the UE according to invention. Note that various embodiments of resource configurators are present in the prior art base stations and UEs for proper synchronization of the UE with the serving BS at the physical layer. However, these resource configurators do not support common CP. Thus, they need new methods for the UE to properly synchronize with the base station and demodulate incoming streams from the base station according to this disclosure.

FIG. 10(a) shows a request-response control messaging sequence wherein Resource Configurator 442 of the UE sends a request to Physical Layer Resource Configurator 441 requesting the resource configuration. In response, Physical Layer Resource Configurator 441 sends configuration parameters associated with all service types it supports according to use of common CP.

FIG. 10(b) shows a request-response control messaging sequence wherein Resource Configurator 442 of the UE sends a request to Physical Layer Resource Configurator 441 requesting the resource configuration for only a specific service. In response, Physical Layer Resource Configurator

441 sends configuration parameters associated with that service type only according to use of common CP.

FIG. 10(c) shows a broadcast control message wherein Physical Layer Resource Configurator 441 sends the resource configuration parameters associated with all service types according to use of common CP.

FIG. 10(d) shows a broadcast control message wherein Physical Layer Resource Configurator 441 sends the common CP duration and selected reference service numerology only. The rest of the needed configuration parameters are pre-configured within each UE's Resource Configurator 442.

One or more of the components illustrated in these Figures can be implemented in specialized circuitry configured to perform the noted functions or one or more of the components can be implemented by one or more processors executing instructions to perform the noted functions. Other embodiments are used without departing from the scope of the present disclosure.

Many of the above-described features and applications can be implemented as software processes that are specified as a set of instructions recorded on a computer readable storage medium (also referred to as computer readable medium). When these instructions are executed by one or more processing unit(s) (e.g., one or more processors, cores of processors, or other processing units), they cause the processing unit(s) to perform the actions indicated in the instructions. Embodiments within the scope of the present disclosure may also include tangible and/or non-transitory computer-readable storage media for carrying or having computer-executable instructions or data structures stored thereon. Such non-transitory computer-readable storage media can be any available media that can be accessed by a general purpose or special purpose computer, including the functional design of any special purpose processor. By way of example, and not limitation, such non-transitory computer-readable media can include flash memory, RANI, ROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code means in the form of computer-executable instructions, data structures, or processor chip design. The computer readable media does not include carrier waves and electronic signals passing wirelessly or over wired connections.

Computer-executable instructions include, for example, instructions and data which cause a general-purpose computer, special purpose computer, or special purpose processing device to perform a certain function or group of functions. Computer-executable instructions also include program modules that are executed by computers in stand-alone or network environments. Generally, program modules include routines, programs, components, data structures, objects, and the functions inherent in the design of special-purpose processors, etc. that perform particular tasks or implement particular abstract data types. Computer-executable instructions, associated data structures, and program modules represent examples of the program code means for executing steps of the methods disclosed herein. The particular sequence of such executable instructions or associated data structures represents examples of corresponding acts for implementing the functions described in such steps.

Processors suitable for the execution of a computer program include, by way of example, both general and special purpose microprocessors, and any one or more processors of any kind of digital computer. Generally, a processor will

receive instructions and data from a read-only memory or a random-access memory or both. The essential elements of a computer are a processor for performing or executing instructions and one or more memory devices for storing instructions and data. Generally, a computer will also include, or be operatively coupled to receive data from or transfer data to, or both, one or more mass storage devices for storing data, e.g., magnetic, magneto-optical disks, or optical disks.

In this specification, the term "software" is meant to include firmware residing in read-only memory or applications stored in magnetic storage or flash storage, for example, a solid-state drive, which can be read into memory for processing by a processor. Also, in some implementations, multiple software technologies can be implemented as sub-parts of a larger program while remaining distinct software technologies. In some implementations, multiple software technologies can also be implemented as separate programs. Finally, any combination of separate programs that together implement a software technology described here is within the scope of the subject technology. In some implementations, the software programs, when installed to operate on one or more electronic systems, define one or more specific machine implementations that execute and perform the operations of the software programs.

A computer program (also known as a program, software, software application, script, or code) can be written in any form of programming language, including compiled or interpreted languages, declarative or procedural languages, and it can be deployed in any form, including as a stand-alone program or as a module, component, subroutine, object, or other unit suitable for use in a computing environment. A computer program may, but need not, correspond to a file in a file system. A program can be stored in a portion of a file that holds other programs or data (e.g., one or more scripts stored in a markup language document), in a single file dedicated to the program in question, or in multiple coordinated files (e.g., files that store one or more modules, sub programs, or portions of code). A computer program can be deployed to be executed on one computer or on multiple computers that are located at one site or distributed across multiple sites and interconnected by a communication network.

These functions described above can be implemented in digital electronic circuitry, in computer software, firmware or hardware. The techniques can be implemented using one or more computer program products. Programmable processors and computers can be included in or packaged as mobile devices. The processes and logic flows can be performed by one or more programmable processors and by one or more programmable logic circuitry. General and special purpose computing devices and storage devices can be interconnected through communication networks.

Some implementations include electronic components, for example microprocessors, storage and memory that store computer program instructions in a machine-readable or computer-readable medium (alternatively referred to as computer-readable storage media, machine-readable media, or machine-readable storage media). Some examples of such computer-readable media include RAM, ROM, read-only compact discs (CD-ROM), recordable compact discs (CD-R), rewritable compact discs (CD-RW), read-only digital versatile discs (e.g., DVD-ROM, dual-layer DVD-ROM), a variety of recordable/rewritable DVDs (e.g., DVD-RAM, DVD-RW, DVD+RW, etc.), flash memory (e.g., SD cards, mini-SD cards, micro-SD cards, etc.), magnetic or solid state hard drives, read-only and recordable Blu-Ray® discs, ultra

density optical discs, any other optical or magnetic media, and floppy disks. The computer-readable media can store a computer program that is executable by at least one processing unit and includes sets of instructions for performing various operations. Examples of computer programs or computer code include machine code, for example is produced by a compiler, and files including higher-level code that are executed by a computer, an electronic component, or a microprocessor using an interpreter.

While the above discussion primarily refers to microprocessor or multi-core processors that execute software, some implementations such as FFT/IFFT and modulation/demodulation operations can be performed by one or more integrated circuits, for example application specific integrated circuits (ASICs) or field programmable gate arrays (FPGAs). In some implementations, such integrated circuits execute instructions that are stored on the circuit itself.

As used in this specification and any claims of this application, the terms “computer readable medium” and “computer readable media” are entirely restricted to tangible, physical objects that store information in a form that is readable by a computer. These terms exclude any wireless signals, wired download signals, and any other ephemeral signals.

What is claimed is:

1. A method executed in a receiver of a user equipment (UE) in a wireless OFDM radio access network (RAN) receiving data over a broadcast radio channel from a base station through M_R receive antennas for a first service type using a lower numerology and a second service type using a higher numerology, wherein a resource configuration for the first service type selected as a reference service type by the base station and communicated by the base station to the UE as using N_0 number of subcarriers, subcarrier spacing of Δf_0 , and cyclic prefix duration of CP_0 and a resource configuration for the second service type having been defined as using N_1 number of subcarriers, subcarrier spacing of Δf_1 and cyclic prefix duration of CP_1 , wherein $L=N_0/N_1$ and $CP_1=CP_0/(N_0/N_1)$, and L is an integer that is positive powers of 2, wherein the receiver performs the steps of:

- a. removing a cyclic prefix with a duration of CP_0 from each time domain symbol generated by the first service type;
- b. removing a cyclic prefix with a duration of CP_0 from each L number of consecutive time domain symbols generated by the second service type;
- c. taking N_0 point FFT of time domain symbols of the first service type obtained in step (a) to create corresponding N_0 frequency domain symbols;
- d. taking L times N_1 point FFT of time domain symbols of the second service type obtained in step (b) to generate corresponding LN_1 frequency domain symbols;
- e. demodulating the N_0 frequency domain symbols generated in step (c) into N_0 bit streams;
- f. demodulating the LN_1 frequency domain symbols generated in step (d) into LN_1 bit streams;

- g. multiplexing bit streams of steps (e) into a serial stream of bits corresponding to the first service type; and
- h. multiplexing bit streams of steps (f) into a serial stream of bits corresponding to the second service type.

2. The method according to claim 1, wherein the lower numerology is numerology zero.

3. The method according to claim 1, wherein the higher numerology is at least numerology one.

4. The method according to claim 1, wherein the UE corresponds to any antenna configurations of MIMO, SISO, SIMO, or MISO.

5. A method executed in a receiver of a user equipment (UE) in a wireless OFDM radio access network receiving data over a broadcast radio channel from a base station through M_R receive antennas for a first service type that is selected as a reference service type by the base station and communicated by the base station to UE, is using a lower numerology and a plurality of other service types using higher numerology than the reference service type, resource configuration for first service type having been communicated by said base station to the UE as using N_0 number of subcarriers, subcarrier spacing of Δf_0 , and cyclic prefix duration of CP_0 and resource configuration for the plurality of other service types been defined as, for service type S_i , using N_i number of subcarriers, subcarrier spacing of Δf_i and cyclic prefix duration of CP_i , wherein $L_i=N_0/N_i$ and $CP_i=CP_0/(N_0/N_i)$, and L_i is an integer that is positive powers of 2, wherein the receiver performs the steps of:

- a. removing a cyclic prefix with a duration of CP_0 from each time domain symbol generated by the first service type;
- b. removing a cyclic prefix with a duration of CP_0 from each L_i number of consecutive time domain symbols generated by each service type S_i ;
- c. taking N_0 point FFT of time domain symbols of the first service type obtained in step (a) to create corresponding N_0 frequency domain symbols;
- d. taking L_i times N_i point FFT of time domain symbols of each service type S_i , obtained in step (b) to generate corresponding $L_i N_i$ frequency domain symbols;
- e. demodulating the N_0 frequency domain symbols generated in step (c) into a bit stream;
- f. demodulating the $L_i N_i$ frequency domain symbols generated in step (d) into a bit stream;
- g. multiplexing bit streams of steps (e) into a serial stream of bits corresponding to the first service type; and
- h. multiplexing bit streams of steps (f) into a serial stream of bits corresponding to each service type S_i .

6. The method according to claim 5, wherein the lower numerology is numerology zero.

7. The method according to claim 5, wherein the higher numerology is at least numerology one.

8. The method according to claim 5, wherein the UE has any antenna configurations of MIMO, SISO, SIMO, or MISO.

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